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a consequentialist approach

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Published in:
International Journal of Sustainable Design

DOI:
10.1504/IJSDES.2019.105401

Publication date:
2019

Document version:
Accepted manuscript

Citation for pulished version (APA):
Haug, A., & Busch, J. (2019). Uncertainties in socially responsible design: a consequentialist approach. *International Journal of Sustainable Design*, *3*(3), 137-151. https://doi.org/10.1504/IJSDES.2019.105401

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Download date: 06. Nov. 2026

Uncertainties in socially responsible design

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Abstract: In recent decades, a plethora of books and papers on socially responsible design has emerged. This literature, however, has not led to the implementation of the solutions needed to solve the environmental and social problems faced by the world today. The lack of initiation of more ambitious projects can to a large extent be explained by the uncertainties associated with the effects of these, as well as the desirability of such effects. More specifically, it can be difficult to get support for highly ambitious projects if there is uncertainty (e.g., in the form of disagreements) about their beneficial effects or the fairness of the distribution of sacrifices needed to implement these. To be able to choose a more ambitious path, we need to better understand the uncertainties associated with socially responsible designs and to reconsider the ethical assumptions guiding our choices. This issue is addressed by defining a framework for understanding uncertainties associated with such projects and by arguing for a consequentialist ethics to govern socially responsible design.

Keywords: socially responsible design; sustainable design; design ethics, uncertainty; risk; product design

1 Introduction

Victor Papanek (1971) was one of the first authors to address the social responsibility of (industrial) designers from an environmentalist perspective in the beginning of the 1970s. Since then a plethora of books and papers related to socially responsible design has emerged. Examples of design philosophies intended to improve the well-being of humans or the environment include design activism, design ethics, ecological design, environmental design, environmentally sustainable design, environmentally conscious design, emotionally durable design, ethical design, green design, nudging approaches, responsible design, social design, sustainable design, triple bottom line and welfare design. However, this literature has not led to the implementation of the solutions needed to solve the environmental and social problems faced by the world today (e.g., Stegall, 2006; Fuad-Luke, 2007; IPCC, 2014a; IPCC, 2014b; Ripple et al., 2017).

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Cite as: Haug, A. and Busch, J. (2019). Uncertainties in socially responsible design: A consequentialist approach. International Journal of Sustainable Design, 3(3), 137-151.

The lack of initiation of more ambitious socially responsible design projects can to a large extent be explained by the uncertainties associated with the effects of these (epistemic uncertainty), as well as the desirability of such effects (ethical uncertainty). Here it should be emphasised that uncertainties are often related to disagreements, rather than not having any answers — in particular, when solutions involve someone having to make sacrifices by paying for such solutions (e.g., through increased income taxes) or by eliminating or minimizing certain activities (e.g., using public transportation instead of driving). In other words, the issue does not just concern the ones developing socially responsible designs, but to a large extent also the ones making the decisions about initiating these.

There are many well-known examples of how epistemic and ethical uncertainty have been major hindrances for action. For example, in relation to wind turbines as a means to reduce greenhouse gas emissions from fossil fuels, there are major debates about the actual effects of this kind of technology (epistemic uncertainty), and if the benefits provided by this technology is worth its costs (ethical uncertainty). On the other hand, if such epistemic and ethical uncertainty was not there, and there was consensus about the effects and that the costs being worth the benefits, there would probably be more wind turbines. Another example is beef cattle production, which involve different perceptions of how much it affects the environment (epistemological uncertainty), and if the enjoyment of beef is worth this effect (ethical uncertainty). Again, if there was consensus about that beef cattle production being a major strain on the environment and the enjoyment of beef not being worth this, there would probably be less beef cattle production. Another example is clothing, for which there is uncertainty about how different fabrics and production methods affect the environment, but in particular, if this is a price worth paying for cheap clothing. More specifically, many companies and consumers, at least indirectly, support the production of environmentally straining clothing, although being aware of these harmful effects.

As the examples above make vivid, the reasons for not implementing the solutions available to address our environmental and social problems can be explained in terms of epistemic and ethical uncertainty. In other words, if we can agree on something being a beneficial solution and the parties who are to make the sacrifices are willing to do so, there would be no reason not to implement it. On the other hand, if we cannot agree on epistemic and ethical matters, then there is uncertainty, which may prevent the implementation of a solution. Thus, to be able to choose a more ambitious path, we need to better understand the uncertainties associated with socially responsible designs and to reconsider the ethical assumptions guiding our choices. On this basis, the paper raises the question:

How can we understand and address uncertainties in socially responsible design?

To address the question in focus, this paper first provides a brief overview of the history of socially responsible design, which is followed by examples from the literature of how socially responsible design initiatives have failed or have at least been questioned with regard to their overall benefits. Next, the paper develops a five-level framework for understanding uncertainties in socially responsible designs. On this basis, the paper argues that a major explanation for the lack of results of socially responsible initiatives in recent decades is that

we seek solutions with minimal sacrifices, and that a consequentialist ethics might help us to gain control over the situation. It is further argued that by focusing on designs that protect certain interests or have a short-term perspective, we take away the attention from the initiatives that could actually make a major difference, and thus may end up doing more harm than good.

2 Literature review

The idea that design practice needs to display social responsibility has a long history. For example, the English artist, writer, and textile designer William Morris (1834-1896), together with the English Arts and Crafts Movement, championed traditional craft production as opposed to division of labour — not only because he believed that this manufacturing method led to superior quality products but also because of the well-paid and honourable employment that he believed craft production provided (Gorman, 2003). In America, designers began to consider social responsibility in the 1930s and 1940s, when, according to Whiteley (1993), the first generation of American industrial designers (including Norman Bel Geddes, Henry Dreyfuss, Walter Dorwin Teague and Raymond Loewy) were trying to justify their practice in terms of contributing to a better world by making products more efficient, easier to operate and more user-friendly. However, some of the downsides of mass production became evident and were criticized in the 1950s — for example, 'planned obsolescence', which was addressed by Vance Packard (1957; 1960).

A radical critique of the practice of design was forwarded in 1971 by Austrian-born US-based designer Victor Papanek (1971) in the book 'Design for the Real World: Human Ecology and Social Change'. This book is widely understood as the seminal text of twentieth-century 'design activism' (Clarke, 2013). In this book, Papanek argued that designers do harm because of their function in supporting over-consumption. Instead, Papanek presented the idea that designers and other creative professionals have a responsibility to cause real change in the world through good design, for example by designing for people's needs rather than their wants. Since then, the topic of 'design activism' has been of growing interest for researchers (e.g., Thorpe, 2008; Fuad-Luke, 2009; Julier, 2013; Markussen, 2013). Generally, design activism is defined as describing the central position of design in relation to: (1) promoting social change, (2) raising awareness about values and beliefs (for example, in relation to sustainability), or (3) questioning the 'negative' role of consumerism in people's everyday life (Markussen, 2013). However, despite the great attention towards socially responsible design approaches, Papanek's agenda is far from being systematically embedded in design education and practice (Boks and Jan Diehl, 2006; Ramirez, 2006).

It was not until the mid-1980s that environmental problems became widely recognized by scientists and intellectuals, and social and environmental concerns were conceptualized on a larger scale and introduced to the public (Er and Kaya, 2008). An important event in this area was the 1987 report 'Our Common Future' by the Brundtland Commission (formally known as United Nations' World Commission on Environment and Development), which positioned 'sustainable development' as a necessity to guide all future human endeavours

(World Commission on Environment and Development, 1987). An oft-quoted definition in the report is that sustainable development concerns "development that meets the needs of the present without compromising the ability of future generations to meet their own needs". Soon after, in 1989, the United Nations Environment Program (UNEP) began the work of identifying and implementing solutions to prevent pollution, and from the early 1990s, industrial designers increasingly focused on cleaner production and began to pay attention to reducing negative impacts along the entire life cycle of a product (Keitsch, 2012). A central element to this development was (and remains) the centrality of the so-called precautionary principle, which advises that lack of evidence about the outcome of an action is not in itself a reason for not taking precautionary measures. An often-cited formulation of the precautionary principle is the formulation found in the 'Rio Declaration on Environment and Development' (or 'Earth Summit Declaration') from 1992: "To protect the environment, the precautionary approach shall be widely applied by states according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation" (United Nations, 1992).

Since the 1980s a number of approaches that deal with environmental or human problems have emerged along with an increased awareness of environmental and humanitarian problems. However, it is widely agreed that the world will continue to face environmental and humanitarian problems, perhaps at an accelerated pace, in the 21st century (Stern, 2006). This message was further emphasized on March 31, 2014, and April 13, 2014, when the Intergovernmental Panel on Climate Change (IPCC) issued its second and third working group reports, respectively. According to these reports, CO2 emissions have been continuously increasing, resulting in climate changes that need to be reacted to quickly and dramatically if humanitarian disasters are to be avoided (IPCC, 2014a; IPCC, 2014b).

In recent decades, various socially responsible designs have been implemented. Many of these, however, have been criticised for having only a minimal effect or even doing more harm than good. To illustrate this, Table 1 shows a small selection of the initiatives whose benefits have been questioned. As compared to the three well-known examples of controversies mentioned in the Introduction section (nuclear power, beef cattle and clothing), here a selection of more surprising examples of controversy is provided to illustrate the complexity of dealing with this matter. It should be emphasised, that it is not the purpose of the authors of this paper to assess whether these criticisms are justified or not. Regardless, such articles may be used by governments, firms and other organisations to argue against such initiatives.

Table 1 Examples of critiques of socially responsible designs

Initiative	*Promoted as*	*Critique of initiative*
Urban agriculture	A means to reduce negative effects of food transportation.	Allocating metropolitan land to agriculture results in lower urban density levels, which results in longer commutes, which is far more energy intensive than food transportation (Glaeser, 2011).
Green	A particularly	Rainwater runoff from green roofs transfers the pollutants seized by

roofs	convenient way of making buildings sustainable.	urban vegetation from the atmosphere to the surrounding environment (Speak, et al., 2014).
Electric vehicles	A less polluting alternative to combustion engine cars.	Electric vehicles do not have a much lower global warming potential, but exhibit the potential for significant increases in human, water and soil toxicity (Hawkins, et al., 2013).
Biofuel	A means to reduce need for fossil fuel and pollution.	The increase in the production of some types of biofuel may threaten biodiversity; some types of biofuels demand more fossil energy to produce than the fossil energy saved by using them (Pimental and Patzek, 2005; Groom et al., 2008).
Bio-degradable products	A means to reduce environmental impact of waste.	Biodegradable products may not be more environmentally friendly when disposed of in landfills because of the methane gas they release when they degrade (greenhouse effect) (Levis and Barlaz, 2011).
Vegetarian products	A means to promote animal welfare and minimize environmental impact.	The switch to vegetarian products leads to an enormous need for palm oil and soya, resulting in massive carbon dioxide emissions, placing animals on the brink of extinction because of the need for plantations, and involving heavy use of pesticides that contaminate ground soil and water (Audsley, et al., 2009).
Vegan clothing	A means to minimize the need for (polluting) animals.	Vegan leather and faux fur creates toxic discharges that contaminate local air, water and soil; plastic-derived products are not fully biodegradable, leading to waste issues (McCutcheon, 2013).
Using less or more natural materials for packaging	A means of becoming more environmentally friendly.	Cutting back too much on packaging or using recycled material can result in damaged products during shipping because of smaller durability, which leads to wasted energy and natural resources; the production process for a paper shopping bag, compared to a standard plastic bag, demands more energy and water, and it releases more greenhouse gases (Porter, 2013).

3 A framework for understanding uncertainties

As shown by the examples in Table 1, it seems that sometimes socially responsible designs fail to produce the intended effects and/or have unforeseen negative side effects. This phenomenon may be explained by the uncertainties related to the effects of any solution. To facilitate an understanding of such uncertainties, three definitions are introduced. These concern: (1) processes producing negative effects, (2) negative effect types, and (3) uncertainties related to effects. On this basis, a framework of uncertainties in socially responsible design is developed.

The first definition focuses on defining the type of processes that can produce negative effects. In this context, a distinction can be made between impacts on humans and the natural environment, and another overall distinction can be made between the phases in the lifecycle of a product, building or service — i.e., ‘before use’ (manufacturing, transport, etc.), ‘during

use' (energy consumption, health issues, etc.), and 'after use' (disposal, recycling, etc.). These two distinctions produce six dimensions that are to be considered to gain a full understanding of the effects of a design. This is illustrated in Figure 1.

Figure 1

The second definition concerns a distinction between focus effect and side effects, and one between direct effects and indirect effects. Focus effects refer to the effects that a solution aims to achieve, while side effects refer to other positive or negative effects. Direct effects refer to the actual effects of a solution, while indirect effects refer to the lost effects of another solution that the chosen solution takes the attention or resources away from. Using these two distinctions, four types of potential negative effects from well-intended socially responsible designs can be derived, as illustrated in Figure 2 and subsequently explained.

Figure 2

Negative direct focus effect refers to situations in which the aim is to achieve an effect in one area, but eventually doing more harm than good in this area. For example, using more fossil energy to produce biofuel than the saved fossil energy as a result of using the biofuel. Negative direct side effects refer to situations in which the aim is to achieve an effect in one area, but by doing so, doing harm in another area. For example, using less packaging material to save resources, but in effect causing more products to be damaged, and thereby wasting other types of resources, as well as causing troubles for individuals and companies. Negative indirect focus effects refer to situations in which the aim is to achieve an effect in one area, but by doing so, blocking for more efficient initiatives in this area. For example, focusing on certain types of alternative energy sources at the expense of more efficient ones. Negative indirect side effects refer to situations in which the aim is to achieve an effect in one area, but by doing so, blocking for initiatives in other areas more in need of attention. For example, every time funds are given to sustainable initiatives addressing one area, at least in principle, it is at the expense of using these funds on initiatives addressing other areas, which some may consider to be more important.

The third definition concerns the notion of risk, which may help us understand why initiatives that produce undesired effects are carried out. According to Hansson (2004), there is a tendency to use the concept of risk to denote any of the following: 1) an unwanted event that may or may not occur; 2) the cause of an unwanted event that may or may not occur; 3) the probability of an unwanted event that may or may not occur; 4) the statistical expectation value of unwanted events that may or may not occur; and 5) the fact that a decision is made under conditions of unknown probabilities. It is not the purpose of this paper to decide between each of these suggested usages that Hansson found in the literature, but it should be highlighted that risk, as it pertains to actions and their impact, can originate both in cases

where the probabilities of different outcomes are known and in cases where they are not. Furthermore, for any action that is carried out, there is always a chance that outcomes different from those expected might occur. Some of the causes for this will be within our range of knowledge (safe range of prediction) and/or control, but some will not be. Because of the probability that can be assigned to each of these outcomes, there is a risk connected with the action, namely the risk of an outcome that was not intended by choosing that action. In worse cases, we may have a grasp of the possible outcomes of an action, but have no idea about the probabilities of the outcomes. In fact, there may be even worse scenarios, namely those where we do not have a full grasp of the possible outcomes of an action that we take. In those cases we do not know whether we have taken every possible harmful outcome into consideration (Sahlin and Persson, 1994). Although it may seem that such actions would be few and far between, it takes little imagination to recognize that in fact many if not all of our actions have uncertain outcomes. The causal chain initiated by any action stretches out in time, i.e., into the future where we (sometimes) have absolutely no clue what the long-term effects of any action might be.

In the context of socially responsible design, obviously, it is impossible to know the exact probabilities of different outcomes in advance, for which reason acting under some degree of uncertainty is unavoidable. Unfortunately, however, such uncertainty is often used as an argument against acting at all, resulting in necessary initiatives remaining unrealized. Furthermore, as mentioned earlier, there is also an ethical aspect, in the sense that efficient solutions often require someone to make certain sacrifices. The problem in this context is that we often cannot agree upon what is fair. To be able to implement more ambitions solutions, we therefore need to address epistemic and ethical uncertainties.

Based on the discussion above, five types of uncertainties may be derived, as illustrated in Figure 3.

Figure 3

The first uncertainty refers to the decision as to which problems are most relevant to solve. Recognizing that we have only limited resources, or that we are willing to invest only a certain amount of resources, a prioritization needs to be made. This type of prioritization is obviously extremely debatable and raises issues such as how to prioritize short-term versus long-term effects and which people deserve attention the most. The risk of this uncertainty is that it may result in decisions that address certain areas at the expense of others, which we may later discover to be in more need of attention. It may also result in favouring present or nearby present (in time) people over future people (Parfit, 1984). The second uncertainty refers to the knowledge as to what the different means are for addressing a problem. The risk of not knowing the full set of possible solution types is that less efficient means may be chosen at the expense of more efficient ones because the decision makers were not aware of their existence and therefore did not investigate. The third uncertainty refers to the direct effect of specific solutions. The risk of this uncertainty is that solutions may be chosen that

fail to produce the expected benefits and/or produce more negative side effects than expected. The fourth uncertainty refers to the indirect effects of specific solutions. The risk of this uncertainty lies in choosing certain solutions at the expense of more efficient ones. The fifth uncertainty refers to the problem of comparing the total effects of different solutions. The problem of this uncertainty is that it involves prioritization as to which positive effects are most desirable and which negative side effects are most tolerable — i.e., there may not be a general consensus about this issue, which can make it difficult to implement such initiatives. Thus, such uncertainties may mean that necessary solutions are not implemented in order to avoid what some perceive as being unacceptable negative effects. However, given that the problems probably cannot be solved without anybody having to pay a price, the emerging question is how we can make such decisions? This is to a large extent an ethical issue, which is addressed in the subsequent section.

4 Addressing ethical uncertainty in socially responsible design

While the three levels of epistemic uncertainty in Figure 3 pertain to the degree to which we are willing and able to investigate the consequences of particular initiatives, the two levels of ethical uncertainty concern the principles that govern our actions. Considering that the fact that we, as the situation is now, could do much more to address the world's environmental and social problems if we were willing to make the sacrifices needed, the ethical issue seems to be the most complex one. To address this issue, this section discusses how consequentialist ethics might help us implement the solutions that are necessary for solving our environmental and humanitarian problems.

According to d'Anjou (2010), ethics in the design disciplines has essentially been articulated around notions that from an overall perspective correspond to Kantian (deontological ethics) and Aristotelian (virtue ethics) perspectives, where the Kantian perspective is the most common in relation to professional codes of ethics and practice. D'Anjou, in turn, argues that Sartre's view of ethics has to be seized as a possible foundation for design ethics. Although the authors of this paper sympathize with d'Anjou that it is unlikely that the solution of the deepest moral dilemmas may be forthcoming, there may more to say, and that given a more consequentialist oriented perspective, we can also make headway towards making better-founded moral choices in design.

Broadly speaking, consequentialist ethics is based on the assumption that the morally right choice is the choice with the best outcome. This stands opposed to an Aristotelian view according to which a morally good choice is made when it is a product of good moral character, and opposed to a Kantian perspective according to which a morally good choice is a choice that is made out of regard for 'the moral law' (based on his idea of the categorical imperative). That we do have coexisting and sometimes conflicting ethical aims is not new, but it is not immediately transparent in either an Aristotelian or a Kantian or a Sartrean view of ethics. On the other hand, consequentialist ethics neatly captures our predicament as moral agents with multiple moral responsibilities and goals.

While consequentialism tells us that the morally best action is the action with the best outcome, this generalized formulation does not define what constitutes a 'best outcome', i.e., what the moral measure should be. In the variety of consequentialist theory, we find different candidates in the history of philosophy. For example, Mill's utilitarian theory advises us to maximize utility, meaning that the goodness of an act becomes relative to its effect on the total happiness: "actions are right in proportion as they tend to promote happiness; wrong as they tend to produce the reverse of happiness" (with 'happiness' roughly defined as pleasure and absence of pain) (Mill, 1861). A related but more recent concept is that of wellbeing, sometimes understood in terms of 'quality of life' (Donaldson et al., 2015). From modern positive psychology we find the vocabulary of preference satisfaction, or simply satisfaction, measured on a numeric scale (Diener et al., 2010). In economics, outcomes are weighed against one another in terms of monetary value. Indeed, one of the major obstacles for modern economics is setting values for such diverse outcomes as environmental catastrophes, species extinctions, the pollution of a lake, and quality human life years.

It is apparent that there are significant challenges inherent to the view in terms of selecting a common measure for ranking outcomes against one another. On the other hand, the view both has the welcome consequence that it allows us to navigate ethically in regard to evaluating the moral goodness of certain choices over others, although those choices have potentially multifarious outcomes with impacts on different aspects of human life and the environment (see, for example, the discussion of discount rates in relation to environmental problems by Broome, 2012). This perspective also seems beneficial in relation to sustainable design. More specifically, when a design is aimed at achieving a particular good, or a good of a particular kind, the idea, generally speaking, is that by implementing one type or novel type of design over another, we are likely to achieve a better outcome with regards to this perceived good than we would be by implementing the other type. The goodness of a design, so to speak, in this picture is a matter of the good consequences that it brings about. As highlighted above, the ethical theory that mirrors these assumptions is consequentialism. According to consequentialism, whether a particular design strategy or design is the morally right one is determined by the actual outcome of that strategy.

As we have been considering so far, any outcome is governed by some degree of risk — be it either in cases where probabilities of outcomes are known in advance or in cases of uncertainty. Thus, the right design strategy to initiate at any given point in time must be decided upon by the expected outcomes of implementation. This expected outcome is a function of the probabilities of possible outcomes. In cases where we can carry out a calculation of this type (where the estimated probabilities are known), we can weigh the expected risk of a design against the expected benefits, granted that we have a way of quantifying outcomes, i.e., in measures of utility, wellbeing, money, etc. In traditional economic fashion, we may conduct a cost-benefit analysis of the situation.

First and foremost, the consequentialist approach requires calculations of good versus bad outcomes, together with a fairly fine-grained ranking of possible outcomes. These types of calculations require much information — a pressing issue is, therefore, the extent to which information of the required type is available. That fact that we are epistemically imperfect

means that we have to rely on educated guesses and that we have got to take precautions when the situation calls for it. When we are interested in socially responsible design, we are interested in designs that are precautionary in nature and which we consider likely to decrease the probability of (or stop) developments towards certain undesirable outcomes or in other ways to contribute to an individual or a common good. In situations such as these, we are (ideally) working with a well-defined value set — one that supports the ethical dimension of our design. Furthermore, we are working on the assumption that we are adequately informed about the causal mechanisms that underlie the processes we wish to manipulate, to make likely the wanted outcomes on the basis of the suggested manipulations.

However, if we keep the distinction pertaining to our epistemic situation separate from the normative requirements that pertain to realizing certain outcomes, we do not obtain the full picture. One thing is the norms that direct us towards realizing certain attractive goals — another thing is the norm that requires us to gather evidence of the (at the point of decision-making) unforeseen consequences of doing so. The adverse effects that any project aimed at a positive outcome may bring about may, on the whole, make the very project one that we should abstain from engaging in. Additionally, cases with the absence of significant evidence, but with a possibility of a very negative outcome, are ones that we perhaps should not engage in, as advocated by the precautionary principle, for example. However, when using the precautionary principle to minimize the probability of a harmful event, precautions should never be excessive. If so, there is a danger that the response to a perceived and plausible threat might be simply to ignore it. Of course, this would not constitute an ethical action. In other words, not being willing to make sacrifices is not a valid excuse for not solving the enormous environmental and humanitarian problems we face. Finally, in cases where we have spent time uncovering the potential risks of a project, we are able to weigh risks and benefits against one another in ways that are more responsible, everything else being equal. The key point here is evidence gathering and weighing in relation to the potential gravity of positive and negative outcomes that projects may have.

Socially responsible design is premised on the assumption that the values, which we have identified and seek to promote, are in fact the right ones to promote. But, of course, it is a rare occurrence that the benefits of a design can be harvested without any negative costs — and not all values can be put on the same scale, for example, the value of living a stress-free life, enjoying the benefits of beautiful surroundings, and the freedom to practice one's religion of choice or the religious value assigned to a plot of land. Furthermore, because the decisions carried out often have consequences way beyond those relating to the decision makers themselves, there are a number of moral questions that immediately present themselves in this connection. If a cost-benefit analysis is carried out, and the benefits of a design is considered to outweigh the possible costs (i.e., the risk of harm), there may still be a problem if harm befalls a number of people who are not involved in the decision to implement the design in question. If people, who are not decision makers, are potentially negatively affected by a design, they only carry the risk of the project without any expected benefit from it, which can pose a moral wrong (Altham, 1984). This is an issue that should be

considered especially in situations in which potential negative indirect consequences, as this type of consequence is the one that we are most likely not to take into consideration.

When we isolate one aim as the one to promote, or isolate a possible outcome as the most important to avoid, we are in serious risk of aiming blindly. However, often when we are interested in enhancing the good for that population overall, at the same time this decreases the goodness for particular individuals of that population, which again raises the issue as to how this should feature in a consequentialist ethics for socially responsible design. Such solutions include making the harmed persons' interests count relatively much, compensating the harmed persons in other ways, or choosing solutions resulting in the people with the highest living standards making the greatest sacrifices. If, on the other hand, we do not allow the goodness of the population to happen at the expense of someone inside or outside of that population, we can ask ourselves if it is possible at all to solve the problems at hand.

The relevance of the discussion above may be illustrated by employing a distinction between costs (of developing and implementing a sustainable solution) and the impact of this solution (in terms of environmental impact), hereby creating four archetypical solution types. These are shown in Figure 4 and subsequently discussed.

Figure 4

As stated in Figure 4, for obvious reasons, solutions with high costs and low impact should be avoided, while solutions with low costs and high impacts should be the first priority. However, it is not always possible to achieve high impact through low-cost solutions; in fact, these situations may be relatively rare. Thus, the question in most situations is if we should choose solutions with low costs and impact or ones with higher costs and impact? As the arrow in Figure 4 symbolises, we argue for the latter. This argument mainly builds upon the observation that in spite the progress made in relation to sustainable designs in recent decades, the environmental problems faced are increasing (IPCC, 2014a; IPCC, 2014b; Ripple et al., 2017). In other words, if to change this negative development, more radical solutions are needed; low cost and low impact solutions are not enough.

5 Conclusion

To address the question, 'how can we understand and address uncertainties in socially responsible designs?', this paper developed three basic definitions. First, by drawing a distinction between effects on humans and the natural environment in combination with three life cycle phases of designs (before use, during use, and after use), six dimensions were derived that need to be considered to gain a full understanding of the effects of a design. Second, the paper introduced a distinction between focus and side effects and a distinction between direct and indirect effects. Using these two distinctions, four types of negative effects stemming from well-intended socially responsible design were defined. Third, the paper made

a distinction between epistemic and ethical uncertainties. On this basis, the paper defined five levels of uncertainty that are central in regard to socially responsible design.

Based on the definitions described above, the paper argued that a significant explanation for the lack of results of socially responsible initiatives in recent decades is that we seek solutions with minimal negative harm to anyone (often economic or lifestyle related), and in this way we end up with solutions that are incapable of solving the environmental and humanitarian problems we face. On this basis, the paper argued that a consequentialist ethics might help us to gain control over the situation and implement solutions that may be undesirable to some people, but which are necessary for solving our environmental and humanitarian problems. The implications for design is a change of focus from on how to make designs that do not to a large extent imply that we need to compromise the way we live today, but also do not have major impacts, to a focus on designs that have major impacts, but also implies certain sacrifices. Obviously, designers do not have the power or the sole responsibility of ensuring that such initiatives are implemented, since this to a larger degree is in the hands of governments and international organisations. However, designers can to some extent affect these through their propositions and visions.

Failure to recognize the messages above may in effect mean that in our quest to make the world a better place we eventually end up doing more harm than good by not doing what is needed in a long-term global perspective because of focusing on designs that protect certain interests or have a short-term perspective; hereby taking the attention and resources away from the initiatives that could actually make a major difference.

References

Altham, J.E.J. (1984) 'Ethics of Risk', *Proceedings of the Aristotelian Society,* Vol. 84, No. 1, pp. 15-29.

Audsley, E., Brander, M., Chatterton, J.C., Murphy-Bokern, D., Webster and C., Williams, A.G. (2009) *How Low Can We Go? An Assessment of Greenhouse Gas Emissions from the UK Food System and the Scope to Reduce Them by 2050*, WWF-UK, Cranfield, UK. http://www.fcrn.org.uk/fcrn/publications/how-low-can-we-go (Accessed 17 March 2017).

Boks, C. and Diehl, J. (2006) 'Integration of sustainability in regular courses: Experiences in industrial design engineering', *Journal of Cleaner Production,* Vol. 14, No. 9-11, pp. 932–939.

Broome, J. (2012) *Climate Matters: Ethics in a Warming World*, Norton, New York.

Clarke, A.J. (2013) '"Actions speak louder": Victor Papanek and the legacy of design activism', *Design and Culture,* Vol. 5, No. 2, pp. 151–168.

D'Anjou, P. (2010) 'Beyond duty and virtue in design ethics', *Design Issues,* Vol. 26, No. 1, pp. 95-105.

Diener, E., Wirtz, D., Tov, W., Kim-Prieto, C., Choi, D.-W., Oishi, S. and Biswas-Diener, R. (2010) 'New well-being measures: Short scales to assess flourishing and positive and negative feelings', *Social Indicators Research*, Vol. 97, No. 2, pp. 143–156.

Donaldson, S.I., Dollwet, M. and Rao, M.A. (2015) 'Happiness, excellence, and optimal human functioning revisited: Examining the peer-reviewed literature linked to positive psychology', *The Journal of Positive Psychology*, Vol. 10, No. 3, pp. 185–195.

Er, Ö. and Kaya, C. (2008) 'Problems or opportunities? Overcoming the mental barrier for

socially responsible design in Turkey', *The Design Journal,* Vol. 11, No. 2, pp. 159–182.
Fuad-Luke, A. (2007) 'Redefining the purpose of (sustainable) design: Enter the design enablers, catalysts in co-design', in Chapman, J. and Gant N.N. (Eds), *Designers, Visionaries and Other Stories*, Earthscan, London, pp. 18-55.
Fuad-Luke, A. (2009) *Design Activism: Beautiful, Strangeness for a Sustainable World*, Routledge, London.
Glaeser, E.L. (2011) 'The Locavore's dilemma: Urban farms do more harm than good to the environment', *The Boston Globe,* 16 June. http://www.boston.com/bostonglobe/editorial_opinion/oped/articles/2011/06/16/the_locavores_dilemma (Accessed 17 March 2017).
Gorman, C. (2003) *The Industrial Design Reader*, Allworth Press, New York, NY.
Groom, M.J, Gray, E. and Townsend, P. (2008) 'Biofuels and biodiversity: Principles for creating better policies for biofuel production', *Conservation Biology, Vol.* 22, No. 3, pp. 602-609.
Hansson, S.O. (2004) 'Philosophical perspectives on risk', *Techne,* Vol. 8, No. 1, pp. 10-35.
Hawkins, T.R., Singh, B., Majeau-Bettez, G. and Strømman, A.H. (2013) 'Comparative environmental life cycle assessment of conventional and electric vehicles', *Journal of Industrial Ecology,* Vol. 17, No. 1, pp. 53-64.
IPCC. (2014a) *Climate Change 2014: Impacts, Adaptation, and Vulnerability*, Cambridge University Press, Cambridge, UK and New York, NY. http://www.ipcc-wg2.gov (Accessed 17 March 2017).
IPCC. (2014b) *Climate Change 2014: Mitigation of Climate Change*. Cambridge University Press: Cambridge, UK and New York, NY. http://www.ipcc-wg2.gov (Accessed 17 March 2017).
Julier, G. (2013) 'From design culture to design activism', *Design and Culture,* Vol. 5, No. 2, pp. 215–236.
Keitsch, M. (2012) 'Sustainable design: A brief appraisal of its main concepts', *Sustainable Development,* Vol. 20, No. 3, pp. 180–188.
Levis, J.W. and Barlaz, M.A. (2011) 'Is biodegradability a desirable attribute for discarded solid waste? Perspectives from a national landfill greenhouse gas inventory model', *Environmental Science & Technology,* Vol. 45, No. 13, pp. 5470–5476.
Markussen, T. (2013) 'The disruptive aesthetics of design activism: Enacting design between art and politics', *Design Issues,* Vol. 29, No. 1, pp. 38–50.
McCutcheon, J. (2013) 'When vegans do more harm than good', *Eluxe Magazine,* 29 August. http://eluxemagazine.com/magazine/vegans-harm (Accessed 17 March 2017).
Mill, J.S. (1861) *Utilitarianism,* Reprint 1979, Hackett, Indianapolis.
Packard, V. (1957) *The Hidden Persuaders*, Penguin Books, London, UK.
Packard, V. (1960) *The Waste Makers*, Penguin Books, Aylesbury, UK.
Papanek, V. (1971) *Design for the Real World: Human Ecology and Social Change*, Pantheon Books, New York.
Parfit, D. (1984) *Reasons and Persons*, Oxford University Press, Oxford.
Pimental, D. and Patzek, T.W. (2005) 'Ethanol production using corn, switchgrass, and wood: Biodiesel production using soybean and sunflower', *Natural Resources Research,* Vol. 14, No. 1, pp. 65-76.
Porter, J. (2013) '3 things you may not know about sustainable packaging, but should', *Entrepreneur*, 31 January. http://www.entrepreneur.com/article/225621 (Accessed 17 March 2017).
Ramirez, M. (2006) 'Sustainability in the education of industrial designers: The case for Australia', *International Journal of Sustainability in Higher Education,* Vol. 7, No. 2, pp. 189–202.

Ripple, W.J., Wolf, C., Newsome, T.M., Galetti, M., Alamgir, M., Crist, E., Mahmoud, M.I. and Laurance, W.F. (2017) 'World scientists' warning to humanity: A second notice', *BioScience* [online] https://doi.org/10.1093/biosci/bix125 (Accessed 26 November 2017).

Sahlin, N.-E. and Persson, J. 1994) 'Epistemic risk: The significance of knowing what one does not know', in Brehmer, B. and Sahlin, N.-E. (Eds.), *Future Risks and Risk Management*, Kluwer Academic Publishers, Dordrecht, pp. 37-62.

Speak, A.F., Rothwell, J.J., Lindley, S.J. and Smith, C.L. (2014) 'Metal and nutrient dynamics on an aged intensive green roof', *Environmental Pollution,* Vol. 184, No. 1, pp. 33-43.

Stegall, N. (2006) 'Designing for sustainability: A philosophy for ecologically intentional design', *Design Issues,* Vol. 22, No. 2, pp. 56–63.

Stern, N. (2006) '*Stern Review: The Economics of Climate Change*', Cambridge University Press, Cambridge, UK.

Thorpe, A. (2008) 'Design as activism: A conceptual tool', in Cipolla, C. and Peruccio, P.P. (Eds.), *Proceedings of the Changing the Change Conference*, Allemandi Conference Press, Turin, pp. 1523-1535.

United Nations. (1992) *Report of the United Nations Conference on Environment and Development, Rio de Janeiro, June 3-4,* United Nations, New York. http://www.un.org/documents/ga/conf151/aconf15126-1annex1.htm (Accessed 17 March 2017).

Whiteley, N. (1993) *Design for Society*, Reaktion Books, London, UK.

World Commission on Environment and Development (1987) '*Our Common Future*', Oxford University Press, Oxford, UK.

Figure 1 Dimensions of socially responsible design

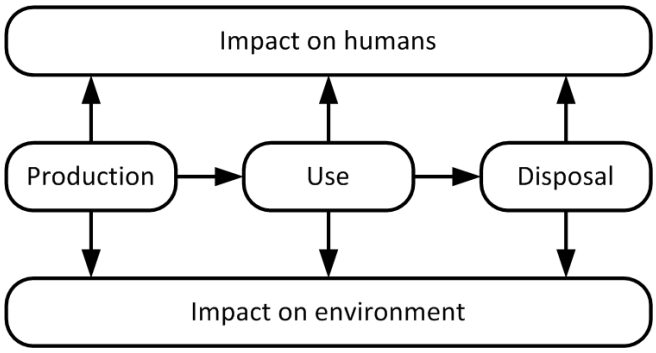


Figure 2 Types of negative effects of socially responsible initiatives

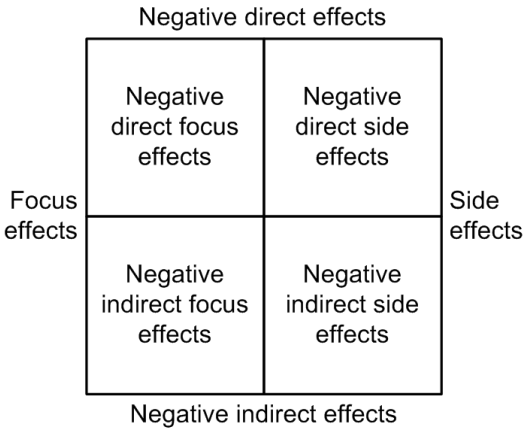


Figure 3 Uncertainties in socially responsible design

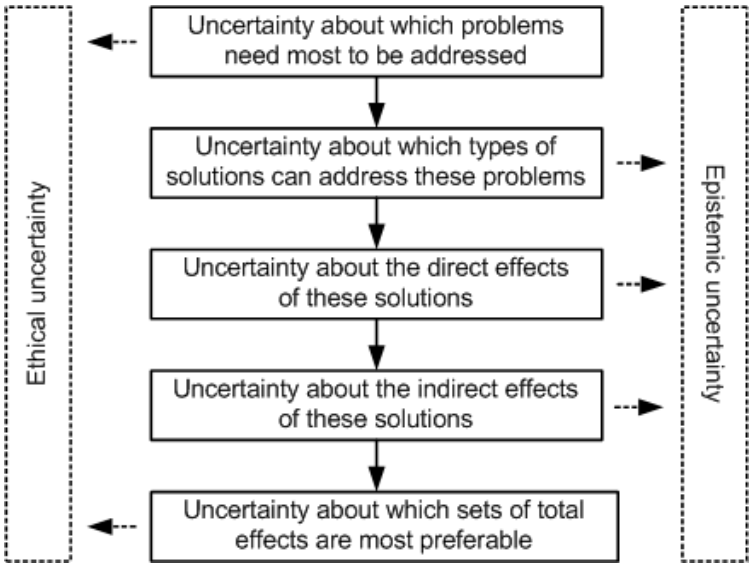


Figure 4 Sustainable design priorities

